**Answers:**

1. $1$
2. $\frac{1}{5}$
3. $\frac{9}{2}$
4. $-1$
5. $\frac{1}{3}$
6. $+\infty$
7. $\frac{(\log a)^2}{2}$
8. $e^2$
9. $0$
10. $1$
11. $1$
12. $0$
13. $2$
14. $+\infty$ for $x > 0$
    $-\infty$ for $x < 0$
    Two sided limit does not exist.
15. $0$
16. $1$
17. $\frac{2}{5}$